Symmetry, Folding, and the Birth of New Solutions

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A graph transformation becomes much easier to understand once the formula is treated as an operation on the domain or the plane. Reflection acts as an involution, absolute value folds one half-space onto another, and intersections of a moving horizontal line with a graph can change only at specific critical values or boundaries. These ideas connect elementary function problems to group actions and the local geometry of bifurcations.[1]

§1.1 A graph center is an involutive symmetry

A graph $y = f(x)$ is invariant under a half-turn about (a, b) precisely when

$$(x, f(x)) \longmapsto (2a - x, 2b - f(x))$$

takes every point of the graph to another point of the graph. Equivalently, the domain is invariant under $x \mapsto 2a - x$, and

$$\boxed{f(a + t) + f(a - t) = 2b.}$$

After the translation $g(t) = f(a + t) - b$, the condition becomes

$$g(-t) = -g(t).$$

Thus a center of symmetry is exactly an odd-function structure after translating the coordinate origin. In particular, the transformation is an involution: applying it twice returns every point to its original position.

Consider the rational function

$$f(x) = \frac{1}{x + 1} + \frac{1}{x + 2}.$$

Its poles are paired around $a = -3/2$. Let $t = x + 3/2$. Then

$$f\left(t - \frac{3}{2}\right) = \frac{1}{t - \frac{1}{2}} + \frac{1}{t + \frac{1}{2}} = \frac{2t}{t^2 - \frac{1}{4}}.$$

The last expression is odd, so the graph has center $(-3/2, 0)$. Notice that a center can be outside the domain: the function is undefined at its two poles, and neither pole needs to be a point of the graph.

The analogous criterion for symmetry about a vertical line $x = a$ is $f(a+t) = f(a-t)$, or evenness after translation. These equations are more reliable than guessing symmetry from a partial sketch.

[1] For the derivative and critical-value viewpoint, see V. I. Arnold, *Catastrophe Theory*, Chapter 1, and the inverse/implicit-function material in MIT 18.02SC: https://ocw.mit.edu/courses/18-02sc-multivariable-calculus-fall-2010/.

§1.2 Absolute values are quotient maps of reflections

The map $q : \mathbb{R} \to [0, \infty)$, $q(x) = |x|$, identifies the reflection pair $\{x, -x\}$. In this sense $[0, \infty)$ is the orbit space of the $\mathbb{Z}/2\mathbb{Z}$ action generated by $x \mapsto -x$.

The graphs of $f(|x|)$ and $|f(x)|$ encode two different foldings:

$$f(|x|) = \begin{cases} f(x), & x \geq 0, \\ f(-x), & x < 0, \end{cases} \qquad |f(x)| = \begin{cases} f(x), & f(x) \geq 0, \\ -f(x), & f(x) < 0. \end{cases}$$

The first folds the *input* across $x = 0$, while the second folds the *output* across $y = 0$. These need not commute.

For $f(x) = x - 1$,

$$|f(|x|)| = ||x| - 1|, \qquad (|f|)(|x|) = ||x| - 1|,$$

which happen to agree. But replacing the input transformation with $|f(x)|$ before folding gives

$$f(|f(x)|) = |x - 1| - 1,$$

a genuinely different function. The order of operations matters because the fold is many-to-one.

A nested absolute value becomes transparent after locating its breakpoints. Set

$$h(x) = \big||x| - 2\big| - 1.$$

Its zeros occur where $\big||x| - 2\big| = 1$, namely at $x \in \{-3, -1, 1, 3\}$. The further folding $H(x) = |h(x)|$ is even, nonnegative, piecewise linear, and has exactly these four zeros. Instead of expanding sixteen cases, one may first analyze $x \geq 0$ on the intervals cut out by $1, 2, 3$, then reflect the result.

§1.3 Intersections change count when transversality fails

Let $F(x, t) = 0$ describe a family of scalar equations depending on a parameter t. If

$$F(x_0, t_0) = 0, \qquad F_x(x_0, t_0) \neq 0,$$

the implicit function theorem gives a unique nearby solution branch $x = x(t)$. Thus a simple root cannot suddenly disappear under a small parameter perturbation.

For roots to appear or merge *inside a smooth region*, one must encounter

$$F(x, t) = 0, \qquad F_x(x, t) = 0.$$

If the graph has corners, jumps, or excluded endpoints, root counts can also change there. These separate mechanisms are visible in a familiar example:

$$|x^2 - 1| = t.$$

Because the left side is nonnegative, there are no roots for $t < 0$. For $t \geq 0$, the equation splits into

$$x^2 = 1 + t, \qquad x^2 = 1 - t,$$

with the second equation admissible only when $t \le 1$. Counting distinct real roots gives

$$\#\{x : |x^2 - 1| = t\} = \begin{cases} 0, & t < 0, \\ 2, & t = 0, \\ 4, & 0 < t < 1, \\ 3, & t = 1, \\ 2, & t > 1. \end{cases}$$

At $t = 0$, the two absolute-value branches meet at the corners $x = \pm 1$. At $t = 1$, the two inner roots coalesce at the smooth maximum $x = 0$ of $1 - x^2$. The root count changes for different geometric reasons at these two levels.

§1.4 The local model is a fold

Near the smooth critical point of the preceding example, write $F(x, t) = 1 - x^2 - t$. At $(0, 1)$,

$$F = 0, \quad F_x = 0, \quad F_t = -1, \quad F_{xx} = -2.$$

The zero set is $t = 1 - x^2$, so for $t < 1$ its two branches are

$$x_\pm(t) = \pm\sqrt{1 - t}.$$

The square-root separation is characteristic of a nondegenerate fold. More generally, suppose $F(x_0, t_0) = F_x(x_0, t_0) = 0$, while F_t and F_{xx} are nonzero. Solving locally for $t = t(x)$ and Taylor expanding gives

$$t - t_0 = -\frac{F_{xx}(x_0, t_0)}{2F_t(x_0, t_0)}(x - x_0)^2 + o((x - x_0)^2).$$

The coefficient determines on which side of t_0 the two nearby solutions exist.

One must still check admissibility: a fold may lie outside the original domain or on a branch discarded by an absolute-value case split. An algebraic multiple root of an auxiliary equation does not automatically describe a valid solution of the original problem.

§1.5 An intersection diagram can be replaced by a proof

For a continuous piecewise monotone function on a compact interval, count the intersections with the level $y = t$ by splitting at all endpoints, corners, and stationary points. On each open monotonicity interval the function takes each intermediate value at most once; the intermediate value theorem determines whether it is taken at all. The only possible changes in the count occur at the finitely many corresponding critical heights.

This procedure explains why symmetry and absolute values are useful *before* computation. Symmetry identifies equal branches, absolute-value foldings create predictable breakpoints, and monotonicity turns a sketch into an exact counting theorem. The higher-level language of involutions and folds merely records the mechanisms already visible in the elementary examples.